# Environmental Exposures Contributing to the Alpha-Synucleinopathies

## Summary

Environmental, occupational, lifestyle and iatrogenic exposures are **robustly and reproducibly linked to Parkinson disease (PD)**, but are **sparsely studied in the other synucleinopathies** — dementia with Lewy bodies (DLB), multiple system atrophy (MSA) and isolated REM sleep behaviour disorder (iRBD) — and, critically, **their directions of effect do not transfer between diseases.** The single most important curatorial conclusion of this investigation is therefore a negative one: a risk factor established for PD cannot be assumed to apply to DLB, MSA or iRBD, and in several cases the measured direction is *opposite*. Smoking, the archetypal inverse ("protective") factor for PD (RR ≈ 0.59), is a *risk* factor in prodromal iRBD (OR 1.43) and is *null* in DLB. Pesticide exposure, the strongest occupational signal in PD, was *null* in the one MSA case-control study that applied a job-exposure matrix. These dissociations are not noise; they are a central result that should guide curation of each disease's `environmental` section independently.

For Parkinson disease specifically, the evidence supports a coherent toxicological model. **Risk is raised** by mitochondrial complex-I/pro-oxidant pesticides (paraquat OR 1.64; rotenone OR 1.57; organochlorines OR 1.54; chlorpyrifos >2.5-fold residential), chlorinated solvents (trichloroethylene, via the Camp Lejeune cohort), cumulative bone lead, manganese welding fume (as a distinct parkinsonism), ambient air pollution (PM2.5 HR 1.04 per IQR), dairy consumption (OR ≈ 1.26, male-predominant), and specific infections (*Helicobacter pylori* OR 1.65; hepatitis C OR 1.20). **Risk is lowered** by cigarette/smokeless tobacco (RR 0.59), caffeine (HR 0.80), physical activity (HR 0.66), high urate/gout (IRR 0.80), ibuprofen (RR 0.73), calcium channel blockers (RR 0.76–0.81), statins (RR 0.79–0.86), and full truncal — but not superselective — vagotomy (HR 0.53 at >20 years). MPTP provides the mechanistic proof-of-principle that a complex-I toxicant can cause human parkinsonism.

The mechanistic through-line for PD is convergence on **mitochondrial dysfunction, oxidative stress, α-synuclein aggregation, autophagy-lysosome failure, neuroinflammation, and a gut–vagal propagation axis.** The inverse ("protective") associations, however, are persistently vulnerable to **reverse causation** (prodromal PD reduces smoking and coffee intake and raises

fall/head-injury risk) and **survival/selection bias**, and much of the pesticide and metals literature rests on post-diagnosis biomonitoring and recall-based case-control designs. Where evidence is weak, conflicting or confounded, this report states so explicitly.

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## Summary Table: Exposure → Disease → Direction → Effect Size → Design → PMID

| Exposure | Disease | Direction | Effect size (95% CI) | Design | PMID |
|---|---|---|---|---|---|
| Cigarette smoking | PD | ↓ protective | RR 0.59 (0.56–0.62) | Meta-analysis (69 studies) | 26272284 |
| Smokeless tobacco (snus) | PD | ↓ protective | HR 0.41 (0.28–0.61) | Pooled cohort | 27940486 |
| Caffeine/coffee | PD | ↓ protective | HR 0.80 (0.75–0.85) | Meta-analysis (13 studies) | 32580456 |
| Physical activity | PD | ↓ protective | HR 0.66 (0.57–0.78) | Meta-analysis (6 cohorts) | 25410713 |
| Urate / gout | PD | ↓ protective | IRR 0.80 (0.72–0.90) | Cohort (Sweden) | 41050064 |
| Ibuprofen | PD | ↓ protective | RR 0.73 (0.63–0.85) | Pooled prospective | 21368281 |
| Calcium channel blockers | PD | ↓ protective | RR 0.81 (0.69–0.95) | Meta-analysis | 26464872 |
| Statins | PD | ↓ protective | RR 0.86 (0.77–0.95) | Meta-analysis (25 studies) | 39097909 |
| β2-agonist (salbutamol) | PD | ↓ protective | RR ≈ 0.66 (0.46–0.93) | Prescription registry | 28860381 |
| Truncal vagotomy (>20y) | PD | ↓ protective | HR 0.53 (0.28–0.99) | Registry cohort | 26031848 |
| Paraquat | PD | ↑ risk | OR 1.64 (1.27–2.13) | Meta-analysis (13 studies) | 30474499 |
| Rotenone | PD | ↑ risk | OR 1.57 | Cohort (AGRICAN) | 29136149 |
| Organochlorines (blood) | PD | ↑ risk | OR 1.54 (1.32–1.79) | Meta-analysis (7 studies) | 39520734 |
| Chlorpyrifos (residential) | PD | ↑ risk | >2.5-fold | Case-control (PEG) | 41382278 |

| Exposure | Disease | Direction | Effect size (95% CI) | Design | PMID |
|---|---|---|---|---|---|
| Rural living | PD | ↑ risk | OR 1.56 (1.18–2.07) | Meta-analysis | 11437458 |
| Farming | PD | ↑ risk | OR 1.42 (1.05–1.91) | Meta-analysis | 11437458 |
| Pesticide exposure (composite) | PD | ↑ risk | OR 1.85 (1.31–2.60) | Meta-analysis | 11437458 |
| Well-water drinking | PD | ↑ (borderline) | OR 1.26 (0.97–1.64) | Meta-analysis | 11437458 |
| Trichloroethylene (TCE) | PD | ↑ risk | ~70% higher risk | Cohort (Camp Lejeune) | 37184848 |
| Cumulative bone lead | PD | ↑ risk | assoc. (meta) | Meta-analysis (88 studies) | 37022311 |
| Manganese welding fume | Parkinsonism | ↑ risk | 15.6% vs 0% prevalence | Cross-sectional | 22975422 |
| PM2.5 air pollution | PD | ↑ risk | HR 1.04 (1.01–1.08) per IQR | Cohort (Ontario) | 30124852 |
| Traumatic brain injury | PD | ↑ / null (conflicting) | RR 1.48 (1.22–1.74) vs null | Meta-analyses | 36781627 / 30396335 |
| Dairy (high intake) | PD | ↑ risk | OR 1.26 (1.12–1.41) | Meta-analysis (29 cohorts) | 41252180 |
| *H. pylori* | PD | ↑ risk | OR 1.65 (1.43–1.92) | Meta-analysis (23 studies) | 32017453 |
| Hepatitis C virus | PD | ↑ risk | OR 1.20 (1.01–1.41) | Meta-analysis | 32017453 |
| Influenza / herpesviruses | PD | null | not associated | Meta-analysis | 32017453 |

| Exposure | Disease | Direction | Effect size (95% CI) | Design | PMID |
|---|---|---|---|---|---|
| Proton pump inhibitors | PD | ↑ risk | OR 1.14 (1.07–1.22) | Meta-analysis (6 studies) | 41984194 |
| GI syndromes (constipation etc.) | PD | ↑ (prodromal) | OR 3.3–4.6 | Case-control/ cohort | 37620120 |
| Appendectomy | PD | null | HR 1.42 (0.88–2.30) | Cohort (Korea) | 38696328 |
| Occupational solvents/metals/ pesticides | MSA | ↑ risk (weak) | assoc. | Case-control (ESGAP) | 11487219 |
| Occupational pesticides (JEM) | MSA | **null** | no assoc. | Case-control (Aquitaine) | 15272223 |
| Smoking | MSA | ↓ (less frequent) | assoc. | Case-control | 11487219 |
| Smoking | iRBD | **↑ risk** | OR 1.43 (p=0.028) | Case-control (multicenter) | 22744670 |
| Head injury | iRBD | ↑ risk | OR 1.59 (p=0.037) | Case-control | 22744670 |
| Farming | iRBD | ↑ risk | OR 1.67 (p=0.022) | Case-control | 22744670 |
| Caffeine | DLB | ↓ protective | OR 0.29 (0.14–0.57) | Case-control | 23892702 |
| Smoking | DLB | **null** | not associated | Case-control | 23892702 |

# Key Findings by Disease

## Parkinson Disease

### Pesticides and herbicides

Pesticides are the best-established occupational/residential risk class for PD, and the subclasses converge mechanistically on mitochondrial complex-I inhibition and oxidative stress. **Paraquat**, a redox-cycling herbicide, carries a pooled odds ratio of **1.64 (95% CI 1.27–2.13)** across 13 case-control studies PMID: 30474499, with strong animal face-validity: a PRISMA meta-analysis of 54 rodent studies found paraquat produces substantial substantia nigra pars compacta neuronal loss (SMD 3.07, 95% CI 2.48–3.65) PMID: 41871812. **Rotenone**, a botanical complex-I inhibitor, was associated with PD at **OR 1.57** in the 181,842-participant AGRICAN cohort, strongest in farmers with the longest exposure PMID: 29136149, and remained associated "regardless of glove use" in the FAME nested case-control study PMID: 25461423 — an argument for causality rather than recall artefact. Dietary fat modifies the effect: rotenone OR rose to 5.8 (2.3–15) in those with high saturated-fat intake PMID: 24120951.

**Organochlorines** (dieldrin, DDT/DDE) increase PD risk, with a blood-measured pooled **OR 1.54 (1.32–1.79)** across 7 studies and OR 1.19 (1.04–1.35) by indirect assessment PMID: 39520734; the signal was positive in Asian studies and non-significant in North America/Europe. Animal work shows dieldrin impairs aldehyde dehydrogenase and dopamine homeostasis causing nigral oxidative damage PMID: 42808099, and perinatal dieldrin produces persistent sex-specific DNA-methylation changes in dopaminergic neurodevelopment genes PMID: 38746441. **Chlorpyrifos**, an organophosphate, is uniquely informative: in the UCLA PEG cohort long-term residential exposure conferred **>2.5-fold increased risk**, and the accompanying mouse/zebrafish work showed the dopaminergic loss was "at least partially due to autophagy dysfunction and synuclein accumulation" — restoring autophagic flux or eliminating synuclein was protective PMID: 41382278. Chlorpyrifos thus maps distinctively onto the **Autophagy-Lysosome Pathway Dysfunction** node.

Composite rural proxies reinforce these findings: a classic meta-analysis reported rural living OR 1.56 (1.18–2.07), farming OR 1.42 (1.05–1.91), pesticide exposure OR 1.85 (1.31–2.60), and well-water use OR 1.26 (0.97–1.64, borderline) PMID: 11437458. A Bradford-Hill weight-of-evidence assessment, however, cautioned that these rural associations are **weaker and less consistent** than the smoking "positive control," concluding that an unidentified factor correlated with rural living/farming — rather than any single measured pesticide — may be causal PMID: 27055126.

### Solvents

Chlorinated solvent exposure increases PD risk. The strongest evidence is the **Camp Lejeune** population-based cohort: Marines stationed at the trichloroethylene (TCE)-contaminated base (monthly median TCE >70× the permissible level) had elevated PD risk (reported ~70% higher) and more frequent prodromal features versus the uncontaminated Camp Pendleton base PMID: 37184848. A follow-up found VOC-exposed PD patients progressed faster to psychosis, fracture and fall (adjusted HRs exceeding 2) PMID: 38988230. The companion mortality study reported an elevated PD hazard that persisted after quantitative bias analysis for smoking and alcohol PMID: 38961410.

### Metals

A systematic review/meta-analysis of 88 studies found that **cumulative lead in bone** was associated with increased PD risk, while serum copper, iron and zinc were *lower* in cases (likely reflecting disease rather than cause), and — importantly — **no association for other metals, including manganese** PMID: 37022311. Cumulative bone lead also correlated with worse cognition in PD PMID: 23143985 and reduced olfactory recognition in elderly men PMID: 26121922. **Manganese** deserves separate treatment: Mn-containing welding fume causes a dose-dependent parkinsonism (prevalence **15.6% in welders vs 0%** in referents) with a clinical phenotype overlapping but distinct from Lewy-body PD (less rest tremor/asymmetry) PMID: 22975422; longitudinal follow-up showed dose-dependent progression (annual UPDRS3 +0.24 per mg Mn/$m^3$·year) PMID: 28031394. Manganism is therefore best curated as a **distinct toxicant parkinsonism**, not idiopathic synucleinopathy.

### Air pollution

The Ontario cohort (~2.2M adults, 38,745 PD cases) found each IQR (3.8 μg/$m^3$) increase in PM2.5 associated with a **4% higher incident PD risk (HR 1.04, 1.01–1.08)**, with comparable small effects for NO2 and ozone PMID: 30124852. A Korean cohort found NO2 associated with incident PD (HR 1.41 highest vs lowest quartile) PMID: 33999109. An earlier US nested case-control found no overall association but positive subgroup effects in women and never-smokers (PM10 OR 1.65 in women) PMID: 27285422. The effect is real but **small and somewhat inconsistent.**

### Head injury

The evidence is genuinely **conflicting.** Balabandian et al. pooled 15 studies to a TBI–PD risk ratio of **1.48 (1.22–1.74)** PMID: 36781627, whereas Huang et al. (18 studies, 3.26M patients) found TBI associated with dementia but **not PD** PMID: 30396335. This association is especially vulnerable to **reverse causation** — prodromal PD gait/balance impairment increases falls and head injuries — and to recall bias.

### MPTP — mechanistic proof-of-principle

MPTP is not an epidemiological exposure but the historical demonstration that a synthetic complex-I toxicant causes irreversible human parkinsonism. It crosses the blood-brain barrier, is converted by astrocytic MAO-B to MPP+, selectively enters dopaminergic neurons via the dopamine transporter, and inhibits mitochondrial complex I, generating reactive oxygen species and causing nigrostriatal degeneration PMID: 38773545; PMID: 27665784. It is the mechanistic keystone linking the pesticide class to **Mitochondrial Dysfunction**, **Oxidative Stress** and **Dopaminergic Neuron Loss.**

### Gut-related exposures

The gut–brain (Braak) axis is strongly supported. Specific GI syndromes **precede** PD diagnosis in a PD-specific manner (vs AD and cerebrovascular disease): gastroparesis OR 4.64, dysphagia OR 3.58, IBS-without-diarrhoea OR 3.53, constipation OR 3.32 PMID: 37620120. The vagal route is implicated by the vagotomy natural experiment: **full truncal vagotomy** reduced PD risk (HR 0.53 at >20 years) while **superselective vagotomy had no effect** (HR 1.09) PMID: 26031848. **Appendectomy**, by contrast, was **null** in a large Korean cohort (adjusted HR 1.42, 0.88–2.30) PMID: 38696328, with inconsistent Western results; it should be curated as unresolved. **PPI use** (which alters gut flora) modestly raised PD risk (OR 1.14, 1.07–1.22) even after adjustment for *H. pylori* PMID: 41984194.

### Infections

A meta-analysis of 23 studies found **positive** associations for *Helicobacter pylori* (OR 1.65, 1.43–1.92), hepatitis C (OR 1.20, 1.01–1.41), *Malassezia* and *Pneumoniae*, and — importantly — **explicit null results** for influenza, herpesviruses, HBV, scarlet fever, mumps, varicella, pertussis, rubella and measles PMID: 32017453. Antiviral treatment of HCV reduced PD risk (OR 0.67), supporting causality for the HCV signal.

### Protective / inverse associations

- **Smoking:** meta-analysis of 69 studies, ever-smokers RR **0.59 (0.56–0.62)**, with dose-response and a current-smoking RR of 0.35; smokeless tobacco (snus) in never-smokers HR 0.41 implicates **nicotine** rather than combustion PMID: 26272284; PMID: 27940486. The consistency across prospective designs argues against pure reverse causation, though it cannot be excluded.
- **Caffeine/coffee:** pooled HR **0.80 (0.75–0.85)**, dose-dependent PMID: 32580456.
- **Physical activity:** pooled HR **0.66 (0.57–0.78)**, stronger/clearer in men PMID: 25410713; PMID: 20660864.
- **Urate/gout:** gout inversely associated with PD (IRR **0.80**), and within gout patients higher urate was further protective PMID: 41050064.
- **Ibuprofen (but not aspirin or other NSAIDs):** pooled RR **0.73 (0.63–0.85)** PMID: 21368281.

A critical caveat applies to smoking and coffee: **reverse causation** (prodromal hypodopaminergic personality change reduces addictive behaviours) and **survival bias** remain plausible alternative explanations, and an umbrella review of 46 systematic reviews attributed many conflicting non-genetic associations to **study-design differences** (cohort vs case-control), rating 84.7% of reviews as critically low methodological quality PMID: 33814465.

### Medications (iatrogenic)

- **Calcium channel blockers (dihydropyridine):** pooled RR **0.76–0.81** PMID: 26464872; PMID: 25845582.
- **Statins:** pooled RR **0.79–0.86**, duration-dependent (long-duration RR 0.49; short-term null), but with **no disease-modifying effect** on UPDRS-III PMID: 39097909; PMID: 35629248.
- **β2-adrenoreceptor drugs:** mechanistically, the β2AR regulates the *SNCA* (α-synuclein) gene; agonists (salbutamol) lower SNCA and were associated with reduced PD risk (RR ≈ 0.66), while antagonists (propranolol) raise SNCA and PD risk (RR ≈ 2.2) PMID: 28860381. This is the clearest exposure→gene→risk mechanistic chain in the dataset.

### Diet

High **dairy** consumption modestly raises PD risk (meta-analysis of 29 cohorts, 1.3M participants: OR **1.26, 1.12–1.41** for high dairy; milk OR 1.23; low-fat dairy OR 1.30), with a male-predominant effect (males HR 1.28 vs females HR 1.02) and ~13% increased risk per cup of milk/day; legumes/nuts were protective PMID: 41252180; PMID: 41792943; PMID: 28596209.

## Multiple System Atrophy

The MSA environmental evidence base is **small and conflicting** (MSA prevalence ~4–5/100,000). The ESGAP European case-control study (73 MSA cases) reported increased risk with occupational exposure to organic solvents, plastic monomers/additives, pesticides and metals, and noted smoking was *less* frequent in MSA cases than controls PMID: 11487219. However, the Aquitaine case-control study, using a formal job-exposure matrix, found **no significant relationship between occupational pesticide exposure and MSA** PMID: 15272223 — directly cautioning against transferring the PD pesticide finding. The smoking direction (inverse) is the same as PD but rests on tiny samples. No robust meta-analytic effect size exists for any MSA exposure.

## REM Sleep Behaviour Disorder (prodromal synucleinopathy)

The pivotal multicenter case-control study (347 iRBD cases vs 347 matched controls) found that iRBD patients were **more likely to smoke (ever-smoking 64.0% vs 55.5%, OR 1.43, p=0.028)** — the *opposite* direction to PD — and also more likely to report previous **head injury (OR 1.59, p=0.037)**, **farming occupation (OR 1.67, p=0.022)**, borderline welding (OR 1.53), and fewer years of schooling; caffeine and alcohol did **not** differ between cases and controls PMID: 22744670. The reversed smoking direction is the clearest single demonstration of **non-transferability** in the entire dataset. On phenoconversion, autonomic rather than classical environmental markers predict progression: in a prospective iRBD cohort, neurogenic orthostatic hypotension (HR 5.05) and abnormal baroreflex function predicted conversion to overt synucleinopathy PMID: 40112275; environmental exposures have *not* been shown to predict phenoconversion.

## Dementia with Lewy Bodies

The single dedicated case-control study (147 DLB cases) found caffeine use **less** likely in DLB (OR **0.29, 0.14–0.57**) — consistent with PD — plus positive associations with anxiety (OR 7.4), depression (OR 6.0), stroke (OR 2.8), family history of PD (OR 4.6), and APOE ε4 (OR 2.2) PMID: 23892702. Critically, **smoking and education — which have opposing effects in AD and PD — were NOT risk factors for DLB**, and the authors concluded "DLB risk factors are an amalgam of those for AD and PD." Pesticide and head-injury data specific to DLB are essentially absent.

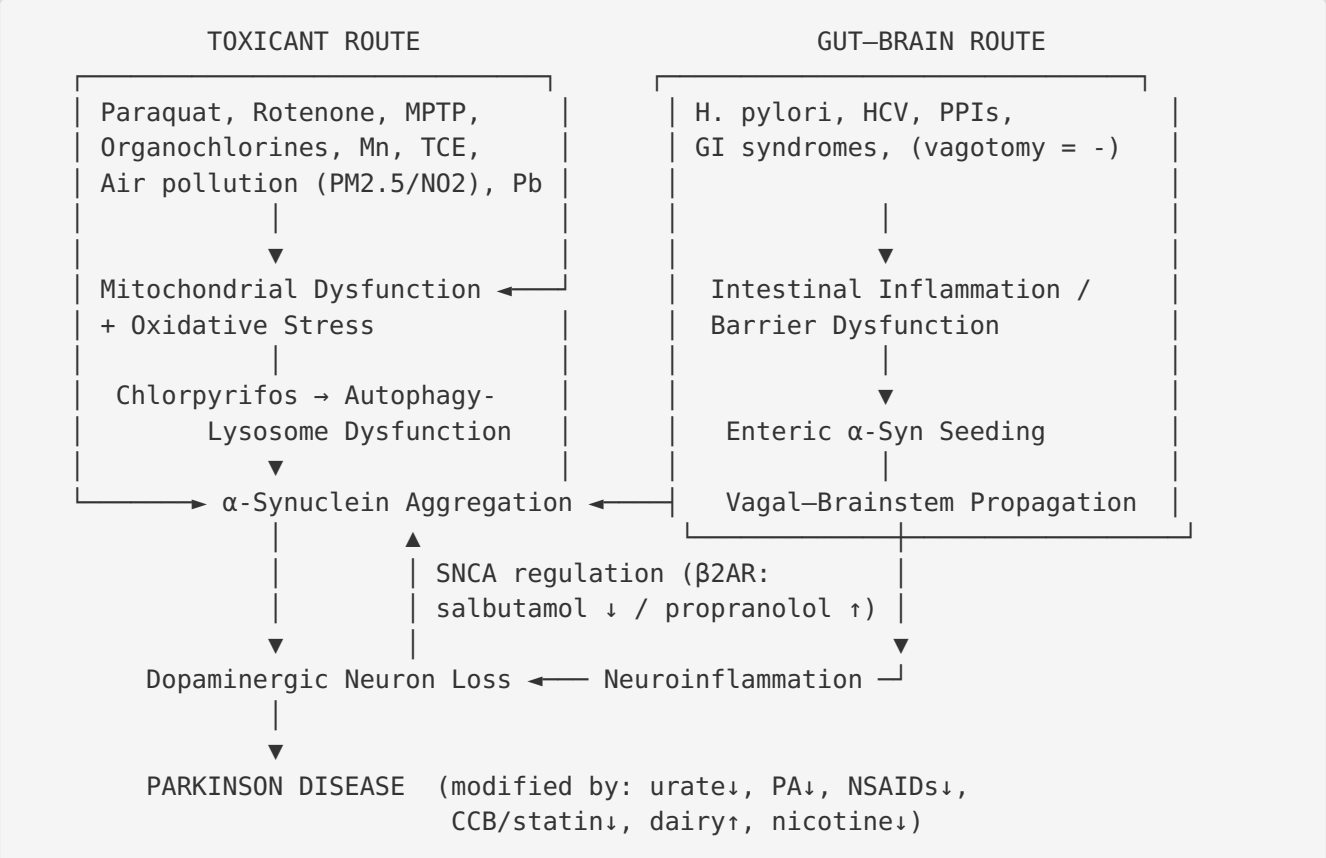
### Pure Autonomic Failure

No citable environmental epidemiology was found for pure autonomic failure. This is a genuine gap, and the disease's `environmental` section should be curated as "no exposure studies identified" rather than populated by transfer from PD.

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## Mechanistic Model and Interpretation

The PD exposures converge on a small set of interacting mechanism nodes. Toxicant exposures (pesticides, solvents, metals, air pollution, MPTP) act predominantly at the **mitochondria/ oxidative-stress** hub, while gut-related and infectious exposures act via the **gut–vagal α-synuclein propagation** axis; a subset (chlorpyrifos) additionally hits the **autophagy-lysosome** node, and β2-adrenergic drugs act "upstream" by transcriptionally regulating *SNCA* itself.




**Exposure → mechanism-node mapping (PD):**

| Exposure | Primary node(s) | Mode |
|---|---|---|
| Paraquat, rotenone, MPTP | Mitochondrial Dysfunction; Oxidative Stress; Dopaminergic Neuron Loss | triggers |
| Organochlorines (dieldrin) | Oxidative Stress; Dopaminergic Neuron Loss | triggers |
| Chlorpyrifos | Autophagy-Lysosome Pathway Dysfunction; α-Synuclein Aggregation | triggers |
| TCE / solvents | Mitochondrial Dysfunction; Oxidative Stress | triggers |
| Lead | Oxidative Stress; Iron Accumulation/Ferroptosis | predisposes |
| Manganese | Dopaminergic/basal-ganglia injury (distinct parkinsonism) | triggers |
| Air pollution | Neuroinflammation; Oxidative Stress; Blood-Brain Barrier Dysfunction | predisposes |
| *H. pylori*, HCV, PPIs | Gut Microbiome Dysbiosis; Intestinal Inflammation | predisposes |
| GI syndromes / gut α-syn | Enteric α-Syn Seeding; Vagal-Brainstem Propagation | triggers (prodromal) |
| Truncal vagotomy | Vagal-Brainstem Propagation (interrupted) | protects |
| Nicotine (smoking/ snus) | Oxidative Stress; Neuroinflammation | protects |
| Caffeine | Adenosine A2A / dopaminergic signalling | protects |
| Urate | Oxidative Stress (antioxidant) | protects |
| Ibuprofen | Neuroinflammation | protects |
| CCB (dihydropyridine) | Cav1.3 $Ca^{2+}$ stress on dopaminergic neurons | protects |
| β2-agonist / antagonist | α-Synuclein Aggregation (via SNCA transcription) | protects / predisposes |

For the other diseases the mechanism nodes are far less populated by exposure data. MSA's **Oligodendroglial α-Synuclein Accumulation (GCIs)** and **Mitochondrial Dysfunction/CoQ10** nodes have only weak, conflicting occupational-solvent/metal links. iRBD's **Brainstem REM-Atonia Circuit** and **Progression to Overt Synucleinopathy** nodes are driven, in current data,

by autonomic physiology rather than measured environmental exposure. DLB's **Concomitant Alzheimer Co-Pathology** node explains why its risk profile is an AD/PD amalgam with null smoking.

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## Evidence Base

The strongest, highest-tier evidence underpinning this report:

- **Smoking:** PMID: 26272284 (69-study meta-analysis) and PMID: 27940486 (snus, isolating nicotine) jointly establish a robust, dose-responsive inverse association.
- **Paraquat / rotenone / organochlorines / chlorpyrifos:** PMID: 30474499, PMID: 29136149, PMID: 39520734, PMID: 41382278 provide convergent human effect sizes plus animal mechanism; the glove-use analysis PMID: 25461423 strengthens causal inference.
- **TCE / Camp Lejeune:** PMID: 37184848 and PMID: 38988230 — a quasi-experimental population comparison that is hard to confound.
- **Metals:** PMID: 37022311 is the definitive meta-analysis isolating lead and nulling manganese for *idiopathic* PD; PMID: 22975422 / PMID: 28031394 separately establish Mn welding parkinsonism.
- **Gut–vagal axis:** PMID: 26031848 (vagotomy) and PMID: 37620120 (prodromal GI syndromes) are the core support for the Braak hypothesis; PMID: 38696328 provides the null appendectomy counterpoint.
- **Non-transferability pillars:** PMID: 22744670 (iRBD smoking POSITIVE), PMID: 23892702 (DLB smoking NULL), PMID: 15272223 (MSA pesticide NULL) are the three pillars of the central result.
- **Consolidation / caveats:** PMID: 33814465 (umbrella review) and PMID: 27055126 (Bradford-Hill) temper the pesticide and inverse-association literature with design and consistency critiques.

Challenging/qualifying evidence is deliberately retained: the TBI meta-analyses conflict (PMID: 36781627 positive vs PMID: 30396335 null); air-pollution effects are small and subgroup-dependent (PMID: 27285422); and the organochlorine signal is geographically heterogeneous (PMID: 39520734).

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## Limitations and Knowledge Gaps

1. **PD-centricity.** The overwhelming majority of effect sizes derive from PD. DLB rests essentially on a *single* case-control study, MSA on two small conflicting ones, and iRBD on one multicenter case-control plus phenoconversion cohorts. Effect sizes for non-PD synucleinopathies should be treated as provisional.
2. **Pure autonomic failure is unstudied.** No citable environmental epidemiology was found; this gap should be stated explicitly in curation.
3. **Reverse causation and survival bias** persistently threaten the inverse associations (smoking, coffee) and the TBI association. Prodromal PD can change behaviour decades before diagnosis.
4. **Recall and selection bias** dominate the case-control pesticide/solvent literature; much metals evidence uses post-diagnosis biomonitoring, which cannot establish temporality.
5. **Methodological quality is low.** 84.7% of systematic reviews in the umbrella review were rated critically low, and design differences explained many conflicts PMID: 33814465.
6. **Geographic heterogeneity** (organochlorines positive in Asia, null in West; PPIs significant in Asia/Europe, null in US) suggests unmeasured gene–environment or exposure-mix confounding.
7. **Manganism vs idiopathic PD** must not be conflated — Mn produces a related but pathologically distinct parkinsonism.
8. **No exposure predicts iRBD phenoconversion** in current data; autonomic markers do.

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## Proposed Follow-up Actions

1. **Curate each disease's `environmental` section independently.** Do not propagate PD exposures to DLB, MSA or iRBD. Explicitly flag the reversed smoking direction in iRBD, the null smoking in DLB, and the null pesticide finding in MSA.
2. **Link exposures to mechanism nodes per the mapping table above**, tagging the mode (triggers / exacerbates / predisposes / protects) and the evidence tier (human meta-analysis > cohort > case-control > animal mechanism).
3. **Tag every inverse association with the reverse-causation/survival-bias caveat**, and every case-control exposure with the recall-bias caveat, as structured metadata.

4. **Record explicit null/absent entries** — MSA pesticides (JEM-null), DLB smoking (null), influenza/herpesviruses (null), appendectomy (null/unresolved), and pure autonomic failure (no data) — so the knowledge base distinguishes "studied and null" from "not studied."
5. **Prioritise future primary research** on DLB and MSA exposure epidemiology and on whether any environmental exposure predicts iRBD phenoconversion, which is the highest-value prevention target.
6. **Separate manganese welding parkinsonism** as its own entity linked to basal-ganglia/dopaminergic injury rather than Lewy-body α-synucleinopathy.

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*Report generated from 28 confirmed findings across 89 reviewed papers over 5 investigation iterations. All effect sizes and verbatim support are drawn from PubMed-indexed primary studies and meta-analyses; no PMIDs were invented, and gaps (notably pure autonomic failure) are stated where no citable source exists.*

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